

Module A (Lec 2–3): Three 1D methods + intuition + convergence order

Order of convergence (Definition 8.1). Given a sequence $\{x^{(k)}\}$ that converges to x^* , we say the *order of convergence* is p ($p \in \mathbb{R}$) if

$$0 < \lim_{k \rightarrow \infty} \frac{\|x^{(k+1)} - x^*\|}{\|x^{(k)} - x^*\|^p} < \infty.$$

Interpretation (Big-O): $\|x^{(k+1)} - x^*\| = O(\|x^{(k)} - x^*\|^p) \Leftrightarrow \exists C > 0, s.t. \forall k \in \mathbb{N}, \|x^{(k+1)} - x^*\| \leq C\|x^{(k)} - x^*\|^p.$

- $p = 1$: **linear** (geometric decay, e.g., Bisection/Golden Section).
- $p = 2$: **quadratic** (digits correct double each step, e.g., Newton).

Bisection (root-finding). Given continuous g and a bracket $[a, b]$ with $g(a)g(b) < 0$:

- Iterate $m = \frac{a+b}{2}$; keep the subinterval where the sign change persists.
- Guarantee: a root stays bracketed; error $\leq \frac{b-a}{2^{k+1}}$ after k steps.
- Convergence: **linear** in interval length, ratio $1/2$ per step.

Golden section (1D minimization). Given unimodal f on $[a, b]$:

- Evaluate at two interior points; discard the side that cannot contain the minimizer.
- Efficiency: after initialization, only *one new* function evaluation per iteration.
- Interval shrink: $(b_k - a_k) \approx 0.618^k(b_0 - a_0)$ (**linear**).

Newton (root-finding / tangents). Given differentiable g :

$$x^{(k+1)} = x^{(k)} - \frac{g(x^{(k)})}{g'(x^{(k)})}.$$

- Intuition: replace g by its tangent line and jump to the tangent's zero.
- Typical local order: **quadratic** when $g'(x^*) \neq 0$; can be higher if derivatives vanish at the root.

Draw Your Plots for the Three Methods:

Practice: 7.8 (Do all three on the same problem)

Root of $g(x) = \frac{e^x - 1}{e^x + 1}$ **Setup facts (use in all parts):** $g(x) = \tanh(x/2)$ and the unique zero is $x^* = 0$.

- (a) **Newton.** Write the Newton iteration for solving $g(x) = 0$ and simplify using $\sinh x = \frac{e^x - e^{-x}}{2}$.

Answer: $g'(x) = \frac{1}{2} \operatorname{sech}^2(x/2)$. Newton: $x^{(k+1)} = x^{(k)} - \frac{g(x^{(k)})}{g'(x^{(k)})} = x^{(k)} - \frac{\tanh(u)}{\frac{1}{2} \operatorname{sech}^2(u)} (u = x^{(k)}/2) = x^{(k)} - 2 \tanh(u) \cosh^2(u) = x^{(k)} - \sinh(x^{(k)})$.

- (b) **Cycling (2-cycle).** Find an equation that an initial condition $x^{(0)} \neq 0$ must satisfy so that Newton cycles: $x^{(0)} = x^{(2)} = x^{(4)} = \dots$.

Answer: Let $N(x) = x - \sinh x$ (Newton map). A sufficient 2-cycle is $N(x^{(0)}) = -x^{(0)}$ (odd symmetry), i.e. $x^{(0)} - \sinh(x^{(0)}) = -x^{(0)} \iff \sinh(x^{(0)}) = 2x^{(0)}$.

- (c) **Convergence region (Newton).** For which $x^{(0)}$ does Newton converge to 0?

Answer: Let $\bar{x} > 0$ be the (unique) solution of $\sinh \bar{x} = 2\bar{x}$. For $0 < x < \bar{x}$, $|N(x)| = \sinh x - x < x$ so magnitudes decrease (sign alternates) $\Rightarrow$ convergence to 0. Thus Newton converges iff $|x^{(0)}| < \bar{x}$; at $|x^{(0)}| = \bar{x}$ it cycles (2-cycle), and for $|x^{(0)}| > \bar{x}$ it diverges.

Module B (Lec 4): First-order condition + Steepest Descent on a quadratic

Unconstrained first-order condition. If f is differentiable and x^* is a local minimizer in $\mathbb{R}^n$, then $\nabla f(x^*) = 0$.
Steepest descent method.

$$x^{(k+1)} = x^{(k)} - \alpha_k \nabla f(x^{(k)}), \quad \text{where } \alpha_k \text{ is chosen by line search on } \phi_k(\alpha) = f(x^{(k)} - \alpha \nabla f(x^{(k)})).$$

Key fact for quadratics: for $f(x) = \frac{1}{2}x^\top Hx + c^\top x + \text{const}$ with $H \succ 0$, exact line search has closed form

$$\alpha_k^* = \frac{\nabla f(x^{(k)})^\top \nabla f(x^{(k)})}{\nabla f(x^{(k)})^\top H \nabla f(x^{(k)})}.$$

Practice: 8.1 (Two iterations + analytic optimum)

Steepest descent from $x^{(0)} = 0$ Minimize

$$f(x_1, x_2) = x_1 + \frac{1}{2}x_2 + \frac{1}{2}x_1^2 + x_2^2 + 3.$$

(a) Compute $\nabla f(x)$ and the analytic minimizer x^* (solve $\nabla f(x) = 0$). **Answer:** $\nabla f(x) = (1 + x_1, \frac{1}{2} + 2x_2)$, so $1 + x_1 = 0$, $\frac{1}{2} + 2x_2 = 0$ gives $x^* = (-1, -\frac{1}{4})$.

(b) Perform two steepest descent iterations with **exact** line search from $x^{(0)} = (0, 0)$: find $x^{(1)}$ and $x^{(2)}$.

Answer: At $x^{(0)} = (0, 0)$, $g^{(0)} = (1, \frac{1}{2})$, $d^{(0)} = -(1, \frac{1}{2})$. Line search on $x(\alpha) = (-\alpha, -\alpha/2)$ gives $\phi(\alpha) = 3 - \frac{5}{4}\alpha + \frac{3}{4}\alpha^2$, so $\alpha_0 = \frac{5}{6}$ and $x^{(1)} = (-\frac{5}{6}, -\frac{5}{12})$. Next $g^{(1)} = (1 + x_1, \frac{1}{2} + 2x_2) = (\frac{1}{6}, -\frac{1}{3})$, $d^{(1)} = -(\frac{1}{6}, -\frac{1}{3}) = (-\frac{1}{6}, \frac{1}{3})$. Here $H = \nabla^2 f = \text{diag}(1, 2)$, so $\alpha_1 = \frac{g^{(1)\top} g^{(1)}}{g^{(1)\top} H g^{(1)}} = \frac{5/36}{1/4} = \frac{5}{9}$. Hence $x^{(2)} = x^{(1)} + \alpha_1 d^{(1)} = (-\frac{25}{27}, -\frac{25}{108})$.

Pitfalls

- Golden section is for **minimization**, bisection is for **root-finding**, Newton's Idea could be for both.
- Newton can have a **small basin of attraction**; cycling can occur (here at $\sinh x = 2x$).
- For steepest descent on quadratics: the direction is easy, but step size matters; use exact line search (or the closed form) when allowed.